

Digital Spectral Geometry and Log-Periodic Saddles

Frontier results for the Thue–Morse, Fabius, inverse-Fabius, and Rvachev systems

Research report prepared for Vladimir Reshetnikov

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Abstract

The Fourier transform of Rvachev’s compactly supported atomic function is the dyadic sinc product

$$\Phi(z) = \prod_{n \geq 1} \operatorname{sinc}\left(\frac{z}{2^n}\right).$$

This report assembles a single manuscript-level chain around the zero divisor, log-periodic endpoint asymptotics, and polynomial reproduction. These topics substantially overlap the repository’s current expositions, frontier compilations, and Lean development; the value claimed here is the coherent presentation, not package-wide novelty or global priority. Regrouping Euler’s products shows that the zero at $2\pi m$ has multiplicity $1 + \nu_2(m)$. Consequently the Fourier-zero spectral zeta function is

$$\mathcal{Z}_\Phi(s) = \frac{\zeta(s)}{(2\pi)^s(1 - 2^{-s})},$$

and the exact zero count is $N(2\pi M) = 2M - s_2(M)$. Thus the binary digit sum—and, by parity, the Thue–Morse sequence—is recoverable from the spectral divisor of the up-function.

The same zeta function yields an exact log-periodic transseries for the moment generating function. Its imaginary poles $2\pi i k / \log 2$ give explicit Fourier coefficients for the periodic term; all ordinary inverse-power corrections vanish because of the trivial zeros of ζ , while the surviving exponentially small sector has coefficients controlled by ν_2 . This makes it possible to write a phase-locked Lambert- W_{-1} saddle equation for the Fabius function at the flat endpoint, derive a refined inverse-Fabius expansion with an explicit periodic modulation, and organize its arbitrary-order saddle corrections.

A further thread develops a sharp dyadic Strang–Fix hierarchy. On the grid $2^{-r}\mathbb{Z}$, canonical Appell-weighted shifts of up reproduce exactly all polynomials of degree at most r , and an explicit Fourier series gives the first unavoidable aliasing defect in degree $r + 1$. Finally, all spectral, digital, cumulant, and transseries constructions are extended to an integer-base family u_b . The report distinguishes proved statements, numerical evidence, and conjectures. A theorem label records a proof in this manuscript only; it does not by itself assert an exact Lean counterpart.

Corpus audit. The package audit was rerun against the materialized `Analysis/FabiusFunction/docs` tree and selected related Lean modules. Its dated inventory, digest method, and overlap register are recorded in `repository_audit.md`. The audit finds substantial prior overlap in each main topic of this report, so no blanket corpus-relative novelty claim is made. It is a source audit, not a literature-priority search or a proof of mathematical equivalence between differently stated results.

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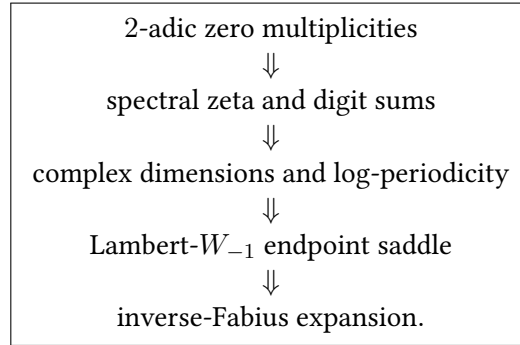
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1 Scope, status, and the main chain of ideas

The repository corpus already develops a large network around the Fabius function, Rvachev’s up, dyadic rational values, the Thue–Morse sequence, q -analogues, infinite sinc products, Bell and Bernoulli polynomials, Lambert W , and polynomial expansions. This manuscript organizes that material around a common analytic object—the *divisor of the entire sinc product*—and uses it to connect four apparently different phenomena:



A second arrow runs from the same zero multiplicities to exact polynomial reproduction on dyadic grids:

$$\text{zero order at reciprocal-lattice frequencies} \iff \text{sharp Strang–Fix degree.}$$

The status conventions are as follows.

- A statement called a theorem, proposition, lemma, or corollary is proved in this manuscript. That label does not mean that the same hypotheses and conclusion have been checked by Lean.
- A numerical table is a high-precision consistency check, never a substitute for proof.
- A conjecture is deliberately stated so that it can be falsified.
- No priority or novelty conclusion should be inferred without a claim-by-claim comparison with the current corpus and external literature.

Current-corpus overlap and Lean boundary. The targeted audit found four especially relevant clusters:

- zero multiplicities, spectral zeta functions, and digital counts in `Frontier_Compilations.tex`, with related formal layers in `DyadicZeroMultiplicity.lean`, `IntegerZeroAnalyticOrder.lean`, and `SpectralZetaWeighted.lean`;
- periodic endpoint corrections and lower-Lambert coordinates in the primary exposition and `Frontier_Compilations.tex`, with related modules `PeriodicCorrection.lean`, the `FabiusLambert*.lean` family, and the `NegativeLaplace*.lean` family;
- Appell synthesis and Strang–Fix sharpness in `Up_Polynomial_Synthesis.tex` and `Frontier_Compilations.tex`, with related modules including `CompositeMeshSharpness.lean` and `RvachevQBinomialFilter.lean`;
- weighted and generalized sinc products in the frontier compilation, with related modules `GeneralizedRvachevEntire.lean`, `GeneralizedZeroDivisor.lean`, and `GeneralizedSincZeta.lean`.

These are topical and declaration-level matches, not assertions that every displayed theorem here has exactly the same formal statement. This report contains no exhaustive theorem-to-declaration crosswalk; accordingly, all of its results remain manuscript/frontier material unless an exact named Lean declaration is supplied separately.

2 Normalization and probabilistic model

2.1 The atomic law

Let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[-1, 1]$, and put

$$X = \sum_{n=1}^{\infty} \frac{U_n}{2^n}. \quad (1)$$

The series converges almost surely and takes values in $[-1, 1]$. Its characteristic and moment generating functions are

$$\Phi(z) = \mathbb{E}e^{izX} = \prod_{n=1}^{\infty} \operatorname{sinc}\left(\frac{z}{2^n}\right), \quad (2)$$

$$M(t) = \mathbb{E}e^{tX} = \Phi(-it) = \prod_{n=1}^{\infty} \frac{\sinh(t/2^n)}{t/2^n}, \quad K(t) = \log M(t). \quad (3)$$

Here $\operatorname{sinc} z = \sin z/z$, with the removable value 1 at the origin.

Proposition 2.1 (Atomic density and the Fabius normalization). *The law of X has an even C^∞ density $u = \operatorname{up}$, supported on $[-1, 1]$, satisfying*

$$u'(x) = 2(u(2x+1) - u(2x-1)). \quad (4)$$

With the standard Fabius normalization,

$$F(x) = u(1-x), \quad 0 \leq x \leq 1. \quad (5)$$

In particular, for $0 \leq x \leq \frac{1}{2}$, $F'(x) = 2F(2x)$.

Proof. The random series gives (2). For $|z|$ large, take $N = \lfloor \log_2 |z| \rfloor - 2$. Since

$$\left| \operatorname{sinc}\left(\frac{z}{2^n}\right) \right| \leq \min\left(1, \frac{2^n}{|z|} e^{|\Im z|/2^n}\right),$$

the product of the first N factors has logarithm

$$-\frac{(\log |z|)^2}{2 \log 2} + \mathcal{O}(\log |z| + |\Im z|).$$

On the real axis this is rapidly decreasing; hence every $z^j \Phi(z)$ is integrable, and Fourier inversion gives a C^∞ density. Its support follows directly from (1).

The product identity

$$\Phi(z) = \operatorname{sinc}(z/2) \Phi(z/2)$$

is the Fourier transform of (4). Evenness follows from the symmetry of each U_n . Formula (5) then gives, for $x \leq \frac{1}{2}$,

$$F'(x) = -u'(1-x) = 2u(1-2x) = 2F(2x),$$

because $u(3-2x) = 0$. □

2.2 Finite splines and the Thue–Morse signs

Let

$$X_N = \sum_{n=1}^N 2^{-n} U_n, \quad a_N = 1 - 2^{-N},$$

and let u_N be its density. Writing $\tau_k = (-1)^{s_2(k)}$, the ordinary inclusion–exclusion formula for a convolution of boxes collapses, in the dyadic case, to

$$u_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{k=0}^{2^N-1} \tau_k \left(x + a_N - \frac{k}{2^{N-1}} \right)_+^{N-1}. \quad (6)$$

This is recalled because it displays the usual, “physical-space” Thue–Morse mechanism. The results below reveal a second mechanism: Thue–Morse parity is also encoded in the *multiplicities of the Fourier zeros*.

The tail is an exact scaled copy:

$$X = X_N + 2^{-N} X', \quad X' \stackrel{d}{=} X, \quad X' \perp X_N. \quad (7)$$

This identity will later explain finite- N q -integers and moment convergence.

3 The Fourier-zero divisor as a digital spectrum

3.1 Canonical product and multiplicities

Euler’s product for the sine gives

$$\operatorname{sinc} w = \prod_{k=1}^{\infty} \left(1 - \frac{w^2}{\pi^2 k^2} \right).$$

Substitution into (2) and a nontrivial regrouping produce the first central result.

Theorem 3.1 (Digital canonical product). *The entire function Φ has the canonical product*

$$\Phi(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{(2\pi m)^2} \right)^{1+\nu_2(m)}. \quad (8)$$

Consequently the nonzero zeros are precisely $\pm 2\pi m$, and the multiplicity at either zero is

$$\operatorname{mult}_{2\pi m} \Phi = 1 + \nu_2(m). \quad (9)$$

The paired genus-zero product in (8) converges locally uniformly.

Proof. Expanding each sinc factor,

$$\Phi(z) = \prod_{n \geq 1} \prod_{k \geq 1} \left(1 - \frac{z^2}{\pi^2 k^2 4^n} \right).$$

Put $m = 2^{n-1}k$. Then $\pi^2 k^2 4^n = (2\pi m)^2$. For a fixed m , the admissible values of n are exactly

$$n = 1, \dots, 1 + \nu_2(m),$$

which proves the multiplicity. Finally,

$$\sum_{m \geq 1} \frac{1 + \nu_2(m)}{m^2} = \zeta(2) \sum_{j \geq 0} 2^{-2j} < \infty,$$

so regrouping is legitimate on compact subsets and the quadratic product converges locally uniformly. $\square$

Remark 3.2. The support and differential equation make u look “dyadic” in physical space. [Theorem 3.1](#) says that its Fourier divisor is dyadic in a much more arithmetic sense: its multiplicity sequence is the ruler sequence $1 + \nu_2(m)$.

3.2 Spectral zeta and exact zero counting

Definition 3.3. For $\Re s > 1$, define the positive-zero spectral zeta function

$$\mathcal{Z}_\Phi(s) = \sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{(2\pi m)^s}. \quad (10)$$

Theorem 3.4 (Closed spectral zeta). *For $\Re s > 1$, and thereafter by meromorphic continuation,*

$$\mathcal{Z}_\Phi(s) = \frac{\zeta(s)}{(2\pi)^s(1 - 2^{-s})}. \quad (11)$$

Besides the pole at $s = 1$, it has the dyadic complex dimensions

$$\chi_k = \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z}, \quad (12)$$

with $k = 0$ interpreted as the pole at the origin.

Proof. Use

$$1 + \nu_2(m) = \sum_{j \geq 0} \mathbf{1}_{2^j | m}.$$

Absolute convergence for $\Re s > 1$ permits the interchange

$$\sum_{m \geq 1} \frac{1 + \nu_2(m)}{m^s} = \sum_{j \geq 0} \sum_{2^j | m} m^{-s} = \zeta(s) \sum_{j \geq 0} 2^{-js} = \frac{\zeta(s)}{1 - 2^{-s}}.$$

The asserted continuation and poles follow from the right-hand side. $\square$

Let $N(R)$ count positive zeros not exceeding R , with multiplicity.

Theorem 3.5 (Exact digit-sum zero count). *For every integer $M \geq 1$,*

$$N(2\pi M) = 2M - s_2(M). \quad (13)$$

More generally,

$$N(R) = 2 \left\lfloor \frac{R}{2\pi} \right\rfloor - s_2 \left(\left\lfloor \frac{R}{2\pi} \right\rfloor \right).$$

Therefore

$$(-1)^{N(2\pi M)} = (-1)^{s_2(M)} = \tau_M. \quad (14)$$

The Thue–Morse sequence is the parity of the cumulative Fourier-zero multiplicity.

Proof. By (9),

$$N(2\pi M) = \sum_{m=1}^M (1 + \nu_2(m)) = M + \nu_2(M!).$$

Legendre’s digit-sum formula gives $\nu_2(M!) = M - s_2(M)$, proving (13). Since $2M$ is even, its parity reduction gives (14). $\square$

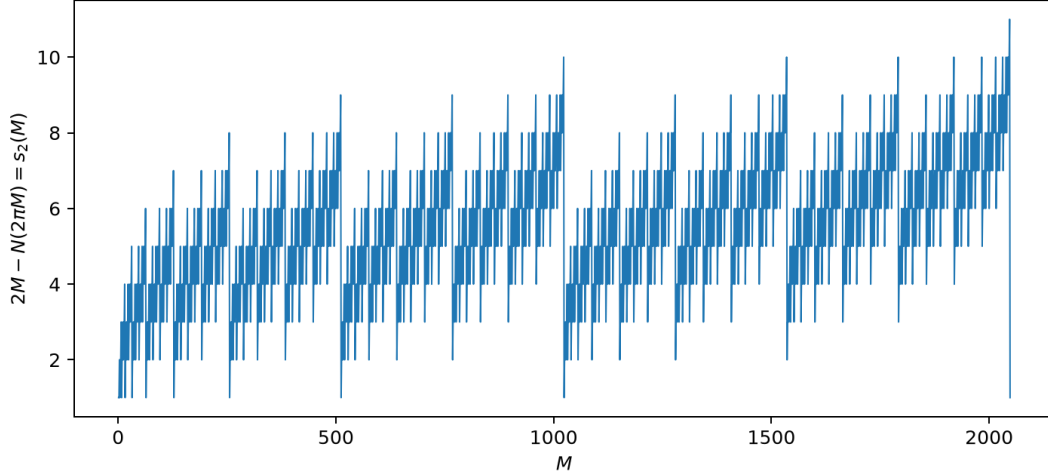


Figure 1: The exact discrepancy in the zero count is the binary digit sum. The graph is not an error term fitted numerically; every plotted point is the identity $2M - N(2\pi M) = s_2(M)$.

3.3 Lambert series and a heat trace

The multiplicity generating function is the lacunary Lambert series

$$\sum_{m \geq 1} (1 + \nu_2(m)) q^m = \sum_{j \geq 0} \frac{q^{2^j}}{1 - q^{2^j}}, \quad |q| < 1. \quad (15)$$

At $q = e^{-2\pi t}$, this becomes a natural heat trace for the positive zero divisor:

$$\mathcal{H}(t) = \sum_{m \geq 1} (1 + \nu_2(m)) e^{-2\pi m t} = \sum_{j \geq 0} \frac{1}{e^{2\pi 2^j t} - 1}. \quad (16)$$

Theorem 3.6 (Heat-trace expansion and complex dimensions). *As $t \downarrow 0$,*

$$\begin{aligned} \mathcal{H}(t) \sim & \frac{1}{\pi t} + \frac{\log t + \gamma}{2 \log 2} - \frac{1}{4} + \mathcal{P}_H(\log_2 t) \\ & + \sum_{r \geq 1} \frac{(2\pi)^{2r-1} B_{2r}}{(2r)! (1 - 2^{2r-1})} t^{2r-1}, \end{aligned} \quad (17)$$

where

$$\mathcal{P}_H(v) = \sum_{k \neq 0} \frac{\Gamma(\chi_k) (2\pi)^{-\chi_k} \zeta(\chi_k)}{\log 2} e^{-2\pi i k v}. \quad (18)$$

The periodic series is real analytic and exponentially convergent.

Proof. Mellin inversion gives, initially on a vertical line $\Re s > 1$,

$$\mathcal{H}(t) = \frac{1}{2\pi i} \int \Gamma(s) \mathcal{Z}_\Phi(s) t^{-s} ds.$$

Move the contour left. The pole at $s = 1$ contributes $1/(\pi t)$. At $s = 0$,

$$\mathcal{Z}_\Phi(s) = -\frac{1}{2(\log 2)s} - \frac{1}{4} + \mathcal{O}(s), \quad \Gamma(s) = \frac{1}{s} - \gamma + \mathcal{O}(s),$$

which gives the logarithmic and constant terms. The poles χ_k , $k \neq 0$, give (18). At $s = -(2r - 1)$, the residue of Γ , together with $\zeta(1 - 2r) = -B_{2r}/(2r)$, gives the displayed Bernoulli term. At negative even integers the trivial zero of ζ cancels the pole of Γ . Stirling's formula for $\Gamma(\chi_k)$ proves exponential convergence of the periodic series. $\square$

4 Spectral-cumulant duality and Bell-polynomial algebra

The canonical product has a direct probabilistic interpretation. Since

$$M(t) = \Phi(-it) = \prod_{m \geq 1} \left(1 + \frac{t^2}{(2\pi m)^2} \right)^{1+\nu_2(m)},$$

we may expand its logarithm at the origin.

Theorem 4.1 (Reciprocal zero power sums are cumulants). *Let κ_j be the cumulants of X . Odd cumulants vanish, and*

$$\kappa_{2r} = (-1)^{r+1} \frac{(2r)!}{r} \mathcal{Z}_\Phi(2r) = \frac{2^{2r-1} B_{2r}}{r(2^{2r} - 1)}. \quad (19)$$

Thus every even cumulant is simultaneously

- (i) a Bernoulli number expression,
- (ii) a reciprocal power sum of Fourier zeros with multiplicity, and
- (iii) the cumulant of the dyadic random series (1).

Proof. From the canonical product,

$$K(t) = \sum_{m \geq 1} (1 + \nu_2(m)) \log \left(1 + \frac{t^2}{(2\pi m)^2} \right) = \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} \mathcal{Z}_\Phi(2r) t^{2r}.$$

Compare with $K(t) = \sum_{r \geq 1} \kappa_{2r} t^{2r} / (2r)!$. Substitution of (11) and Euler's formula for $\zeta(2r)$ yields the Bernoulli form. $\square$

The first cumulants are

$$\kappa_2 = \frac{1}{9}, \quad \kappa_4 = -\frac{2}{225}, \quad \kappa_6 = \frac{16}{3969}, \quad \kappa_8 = -\frac{16}{3825}.$$

The moments are complete Bell polynomials:

$$m_{2n} := \mathbb{E}X^{2n} = (2n)! \sum_{\substack{j_1+2j_2+\dots+nj_n=n \\ j_r \geq 0}} \prod_{r=1}^n \frac{1}{j_r!} \left(\frac{\kappa_{2r}}{(2r)!} \right)^{j_r}. \quad (20)$$

For example,

$$m_2 = \frac{1}{9}, \quad m_4 = \frac{19}{675}, \quad m_6 = \frac{583}{59535}.$$

4.1 Finite products and q -integers

For the finite product

$$\Phi_N(z) = \prod_{n=1}^N \text{sinc}(z/2^n),$$

the zero at $2\pi m$ has multiplicity

$$\min(N, 1 + \nu_2(m)) = \sum_{j=0}^{N-1} \mathbf{1}_{2^j | m}.$$

Hence its spectral zeta is

$$\mathcal{Z}_{\Phi,N}(s) = (2\pi)^{-s} \zeta(s) \frac{1 - 2^{-Ns}}{1 - 2^{-s}} = (2\pi)^{-s} \zeta(s) [N]_{2^{-s}}, \quad (21)$$

where $[N]_q = (1 - q^N)/(1 - q)$. This is a direct q -analogue in the zero spectrum, not merely a formal resemblance.

The finite cumulants consequently satisfy

$$\kappa_{2r,N} = \kappa_{2r}(1 - 2^{-2rN}). \quad (22)$$

Proposition 4.2 (First finite-spline moment correction). *For fixed $n \geq 1$,*

$$m_{2n,N} = m_{2n} - \binom{2n}{2} \kappa_2 m_{2n-2} 2^{-2N} + \mathcal{O}(2^{-4N}). \quad (23)$$

A full expansion in powers of 2^{-2N} is obtained by substituting (22) into the Bell polynomial (20).

Proof. For complete Bell polynomials,

$$\frac{\partial m_j}{\partial \kappa_\ell} = \binom{j}{\ell} m_{j-\ell}.$$

The only cumulant perturbation of order 2^{-2N} is $-\kappa_2 2^{-2N}$; all other direct perturbations and all quadratic terms are $\mathcal{O}(2^{-4N})$. $\square$

4.2 A Fourier–Bessel bridge to Legendre coefficients

Write the even Legendre expansion

$$u(x) = \sum_{n=0}^{\infty} a_{2n} P_{2n}(x), \quad a_{2n} = \frac{4n+1}{2} \mathbb{E} P_{2n}(X). \quad (24)$$

Proposition 4.3 (Sinc-product representation of Legendre coefficients). *Let j_n denote the spherical Bessel function. Then*

$$a_{2n} = \frac{(4n+1)(-1)^n}{\pi} \int_0^\infty \Phi(t) j_{2n}(t) dt. \quad (25)$$

Moreover,

$$|a_{2n}| \leq \exp\left(-\frac{(\log n)^2}{2 \log 2} + \mathcal{O}(\log n)\right). \quad (26)$$

Proof. Fourier inversion and

$$\int_{-1}^1 e^{-itx} P_m(x) dx = 2(-i)^m j_m(t)$$

give (25). For the estimate, split the integral at a fixed fraction of $2n$. In the subturning region, standard bounds for j_{2n} are exponentially small in n . In the remaining region,

$$|\Phi(t)| \leq \exp\left(-\frac{(\log |t|)^2}{2 \log 2} + \mathcal{O}(\log |t|)\right),$$

as in the proof of [proposition 2.1](#); the integrable Bessel envelope and a dyadic decomposition of the tail then give (26). $\square$

Remark 4.4. The bound is an envelope. Because Φ has exact high-multiplicity zeros, individual Legendre coefficients may be substantially smaller through oscillatory cancellation. Determining their sharp logarithmic asymptotics is one of the research problems in [section 12](#).

5 An exact log-periodic transseries for $K(t)$

Put

$$f(x) = \log \frac{\sinh x}{x}, \quad K(t) = \sum_{n \geq 1} f(t/2^n), \quad a = \log 2.$$

The usual statement that $K(t)$ has a log-squared asymptotic conceals two different effects: a real-analytic periodic fluctuation of $\log_2 t$, and a separate exponentially small digital sector. Both can be written exactly.

Theorem 5.1 (Exact split-sum transseries). *For $t \geq 1$, let*

$$q = \log t, \quad \theta = \{\log_2 t\}.$$

Then

$$K(t) = t - \frac{q^2}{2a} - \frac{q}{2} + \Psi(\theta) + R(t), \quad (27)$$

where Ψ is the 1-periodic, real-analytic function

$$\begin{aligned} \Psi(\theta) = & -2^\theta + \frac{a}{2}(\theta^2 + \theta) \\ & + \sum_{j=0}^{\infty} \log(1 - e^{-2^{j+\theta+1}}) + \sum_{k=1}^{\infty} \log\left(\frac{\sinh(2^{\theta-k})}{2^{\theta-k}}\right), \quad 0 \leq \theta < 1, \end{aligned} \quad (28)$$

and

$$R(t) = - \sum_{\ell=0}^{\infty} \log(1 - e^{-2^{\ell+1}t}) \quad (29)$$

$$= \sum_{n=1}^{\infty} \frac{2^{\nu_2(n)+1} - 1}{n} e^{-2nt}. \quad (30)$$

In particular, there are no algebraic inverse powers of t :

$$R(t) = e^{-2t} + \frac{3}{2}e^{-4t} + \mathcal{O}(e^{-6t}).$$

Proof. Write $\log_2 t = N + \theta$, with $N = \lfloor \log_2 t \rfloor$. For $n \leq N$,

$$f(t/2^n) = \frac{t}{2^n} - \log\left(\frac{2t}{2^n}\right) + \log(1 - e^{-2t/2^n}).$$

The first two finite sums are

$$\begin{aligned} \sum_{n=1}^N \frac{t}{2^n} &= t - 2^\theta, \\ - \sum_{n=1}^N \log\left(\frac{2t}{2^n}\right) &= -\frac{q^2}{2a} - \frac{q}{2} + \frac{a}{2}(\theta^2 + \theta). \end{aligned}$$

Reindex the exponential sum by $j = N - n$, and the small-argument tail by $k = n - N$. Extending the finite j -sum to infinity creates exactly the remainder (29). The expression is unchanged when θ is increased by 1 and the two sums are reindexed, proving periodicity.

Finally, expand each logarithm in (29). In a term with exponent $2nt$, write $n = m2^\ell$. Summing $1/m = 2^\ell/n$ over $0 \leq \ell \leq \nu_2(n)$ gives $(2^{\nu_2(n)+1} - 1)/n$, proving (30). $\square$

5.1 Explicit Fourier coefficients from the spectral zeta

The periodic function is extraordinarily flat, but not constant.

Theorem 5.2 (Fourier series of the periodic fluctuation). *Let*

$$\Psi(\theta) = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k \theta}.$$

With γ_1 denoting the first Stieltjes constant in the convention

$$\zeta(s) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n \gamma_n}{n!} (s-1)^n,$$

the coefficients are

$$c_0 = \frac{\gamma_1 + \gamma^2/2 - \pi^2/12}{\log 2} - \frac{\log 2}{12}, \quad (31)$$

$$c_k = \frac{\pi(2\pi)^{-\chi_k} \zeta(\chi_k)}{(\log 2) \chi_k \sin(\pi \chi_k/2)}, \quad \chi_k = \frac{2\pi i k}{\log 2}, \quad k \neq 0. \quad (32)$$

The series and all of its derivatives converge absolutely. In particular,

$$c_k = \mathcal{O}\left(|k|^A e^{-\pi^2 |k|/\log 2}\right)$$

for some fixed A .

Proof. The canonical product gives

$$K(t) = \sum_{m \geq 1} (1 + \nu_2(m)) \log \left(1 + \frac{t^2}{(2\pi m)^2} \right).$$

For $1/2 < c < 1$,

$$K(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi}{s \sin \pi s} \mathcal{Z}_\Phi(2s) t^{2s} ds.$$

Moving the contour left crosses $s = 1/2$, the origin, and $s = \chi_k/2$. The first pole contributes t . At the origin,

$$\mathcal{Z}_\Phi(2s) = -\frac{1}{4as} - \frac{1}{4} + s \left(-\frac{a}{12} + \frac{\zeta''(0)}{a} + \frac{\log^2(2\pi)}{2a} \right) + \mathcal{O}(s^2),$$

while $\pi/(s \sin \pi s) = s^{-2} + \pi^2/6 + \dots$. Taking the residue after multiplication by t^{2s} gives

$$-\frac{(\log t)^2}{2a} - \frac{\log t}{2} + c_0,$$

where the standard identity

$$\zeta''(0) = \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{24} - \frac{1}{2} \log^2(2\pi)$$

yields (31). The residue at $s = \chi_k/2$ is (32). At every negative integer, the pole of $\pi/(s \sin \pi s)$ is cancelled by a trivial zero of $\zeta(2s)$; this is the Mellin explanation for the absence of inverse powers in theorem 5.1. The exponential estimate follows from $\sin(\pi \chi_k/2) = i \sinh(\pi^2 k/\log 2)$ and standard vertical-line bounds for ζ . $\square$

$$\begin{aligned} c_0 &= -1.109045365434186715431, & |c_1| &= 1.06271162519e - 6, \\ \arg c_1 &= -0.779506592173. \end{aligned} \quad (33)$$

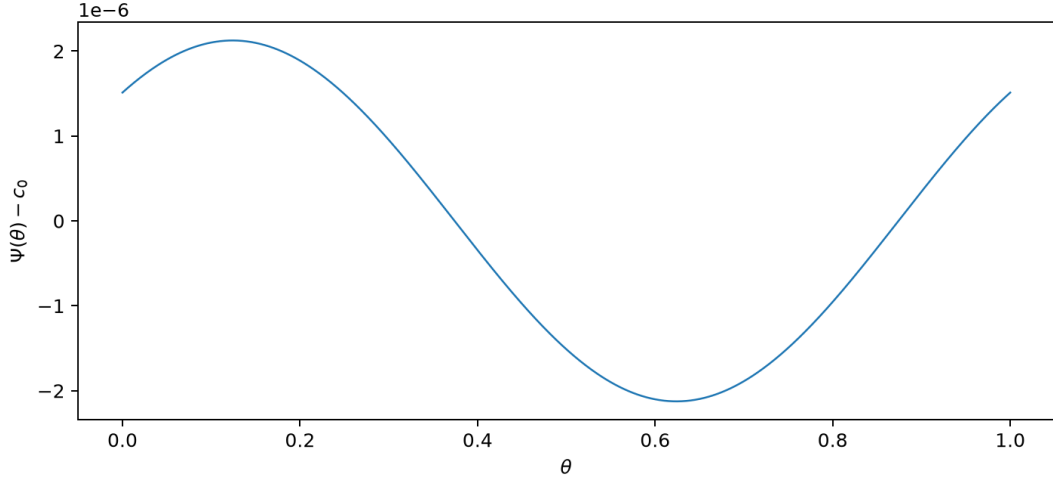


Figure 2: The centered periodic fluctuation $\Psi(\theta) - c_0$. Its small amplitude is a consequence of the factor $\sinh(\pi^2 k / \log 2)$ in (32), not numerical roundoff.

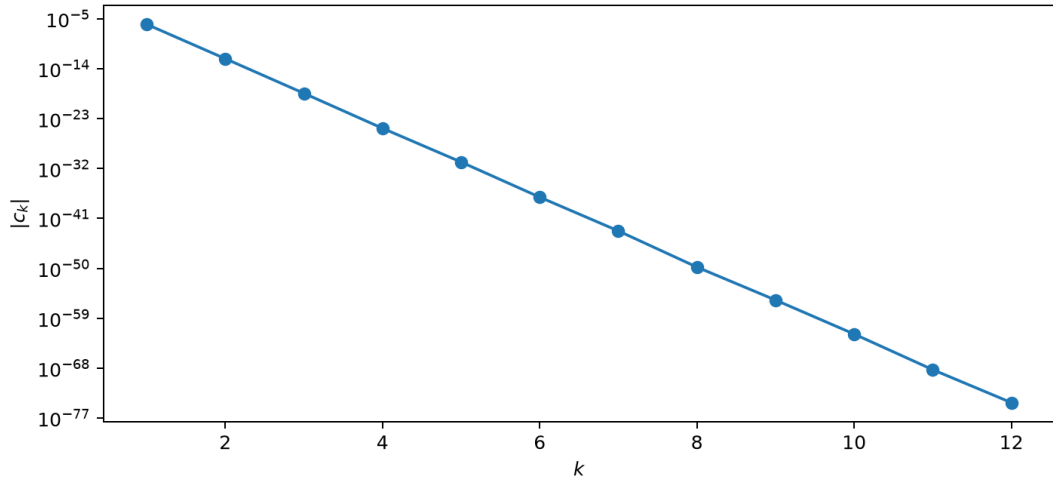


Figure 3: Exponential decay of the Fourier coefficients c_k . The first harmonic already resolves the visible periodic oscillation.

Table 1: High-precision verification of the exact decomposition $K(t) = t - (\log t)^2 / (2 \log 2) - (\log t) / 2 + \Psi(\log_2 t) + R(t)$.

t	$K(t)$ from the defining product	absolute discrepancy
2.3	0.284279799605844922	$4.9983e - 17$
10.0	3.91515171832612015	$1.612e - 16$
100.0	81.2903232099920691	$5.4506e - 16$
1000.0	961.016478652497599	$1.1848e - 15$

Table 2: The split-sum definition of Ψ versus its Fourier series (eight positive harmonics).

θ	$\Psi(\theta)$	absolute discrepancy
0.0	-1.10904385370402886	$3.3242e - 17$
0.125	-1.10904324004899393	$3.161e - 17$
0.25	-1.10904387141412452	$2.9616e - 17$
0.375	-1.10904537795502636	$2.726e - 17$
0.5	-1.10904687716304526	$2.4542e - 17$
0.625	-1.10904749082172016	$2.1463e - 17$
0.75	-1.10904685945554812	$1.8022e - 17$
0.875	-1.10904535291100633	$1.4218e - 17$

Table 3: Internal consistency check for the explicit inverse expansion. Here $q = \log t$ is prescribed, $x = 1 - K'(t)$, and Y is obtained from the exact cumulant function together with the leading saddle prefactor.

q	x	Y	$q_{\text{app}} - q$
8.0	0.004039496674	33.3330382604	0.0054889982
12.0	0.0001094429121	83.4873806064	0.0022249978
16.0	$2.653930052e - 6$	156.658529975	0.0011900322
20.0	$6.050285961e - 8$	252.877624817	0.00073672587

6 The endpoint saddle and a phase-locked Lambert equation

Because $F(x) = u(1 - x)$, the flat endpoint $x \downarrow 0$ is a right-edge large-deviation problem for X . The cumulant function K is strictly convex, $K'(0) = 0$, and $K'(t) \uparrow 1$. Hence, for every $0 < x < 1$, there is a unique $t = t(x) > 0$ such that

$$K'(t) = 1 - x. \quad (34)$$

Set

$$q = \log t, \quad \theta = \{q/a\}, \quad p(\theta) = \Psi'(\theta).$$

Differentiating (27) gives the exact saddle equation

$$xt = \frac{q}{a} + \frac{1}{2} - \frac{p(\theta)}{a} - tR'(t). \quad (35)$$

The exponentially small term is explicit from (29).

Theorem 6.1 (Phase-locked Lambert- W_{-1} form). *After omission of the exponentially small R' -sector, (35) is equivalent to*

$$t = -\frac{1}{ax} W_{-1}(-ax \exp\{-a/2 + p(\log_2 t)\}). \quad (36)$$

Its zeroth approximation,

$$t_0(x) = -\frac{1}{ax} W_{-1}\left(-\frac{ax}{\sqrt{2}}\right), \quad (37)$$

has relative error $\mathcal{O}(\|\Psi'\|_\infty / \log(1/x))$, apart from the explicitly smaller R' -sector. Iteration of the right-hand side of (36) is contractive for sufficiently small x .

Proof. Let $y = ax t$. Without R' , (35) reads

$$y = \log t + \frac{a}{2} - p(\log_2 t) = \log \frac{y}{ax} + \frac{a}{2} - p(\log_2 t).$$

Therefore

$$ye^{-y} = ax e^{-a/2 + p(\log_2 t)}.$$

The relevant solution has $y \rightarrow \infty$, so it is the W_{-1} -branch, proving (36). The derivative of the induced fixed-point map contains a factor $1/(y-1)$, while Ψ'' is bounded; hence it is a contraction when y is large. Linearization in the bounded phase perturbation gives the error estimate. $\square$

Remark 6.2. The Lambert approximation is much better than its formal order might suggest, because Ψ' is exponentially small in the complex-dimension spacing. Nevertheless, retaining it is conceptually important: it is the first explicit mechanism by which the dyadic spectral divisor changes the inverse-Fabius phase.

6.1 Arbitrary-order saddle expansion

Let $t = t(x)$ solve (34). Bilateral Laplace inversion along the vertical line through t gives

$$F(x) = \frac{e^{K(t)-t(1-x)}}{2\pi} \int_{-\infty}^{\infty} \exp(K(t+iy) - K(t) - iyK'(t)) dy.$$

The tilted variance is $K''(t)$. Put

$$\lambda_j(t) = \frac{K^{(j)}(t)}{j! K''(t)^{j/2}}, \quad j \geq 3.$$

Formal Gaussian expansion of (6.1) gives universal polynomials in the λ_j .

Theorem 6.3 (All-order edge expansion). *As $x \downarrow 0$, equivalently $q = \log t \rightarrow \infty$,*

$$F(x) = \frac{\exp\{K(t) - t(1-x)\}}{\sqrt{2\pi K''(t)}} \left[1 + \frac{K^{(4)}(t)}{8K''(t)^2} - \frac{5K^{(3)}(t)^2}{24K''(t)^3} + \mathcal{O}(q^{-2}) \right]. \quad (38)$$

More generally, truncation after weighted Gaussian degree $2m$ has relative error $\mathcal{O}(q^{-m})$. Every coefficient is generated by

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-z^2/2} \exp\left(\sum_{j \geq 3} \lambda_j(t)(iz)^j\right) dz, \quad (39)$$

expanded formally and evaluated using $\mathbb{E}Z^{2r} = (2r-1)!!$.

Proof. Under the exponential tilt with parameter t , X is still a sum of independent bounded variables. The number of summands contributing non-negligibly is asymptotic to $\log_2 t$. Direct differentiation of (27) shows

$$K''(t) = \frac{q/a + \mathcal{O}(1)}{t^2}, \quad \lambda_j(t) = \mathcal{O}(q^{1-j/2}).$$

After the scaling $y = z/\sqrt{K''(t)}$, the central part of (6.1) has the expansion (39); odd Gaussian moments vanish. The first nonzero terms are the quartic term and half the square of the cubic term, which yield (38). Boundedness of the summands under the tilt gives a uniform characteristic-function gap outside the central region, completing the standard saddlepoint remainder estimate. $\square$

7 A refined asymptotic for the inverse Fabius function

The preceding formulas can be inverted explicitly one order beyond the usual square-root logarithm, while retaining the spectral phase.

Let

$$Y = \log \frac{1}{y}, \quad Q = \sqrt{2aY}, \quad c = 1 + \frac{a}{2},$$

and define

$$C(\theta) = -\frac{1}{2} - \Psi(\theta) + \frac{\Psi'(\theta)}{a} + \frac{1}{2} \log \frac{2\pi}{a}. \quad (40)$$

Theorem 7.1 (Log-periodically modulated inverse-Fabius expansion). *As $y \downarrow 0$, define the leading phase*

$$\theta_0 = \left\{ \frac{Q + c}{a} \right\}.$$

Then

$$q(y) = Q + c + \frac{-\frac{a}{2} \log Q + \frac{c^2}{2} - aC(\theta_0)}{Q} + \mathcal{O}\left(\frac{(\log Q)^2}{Q^2}\right), \quad (41)$$

$$F^{-1}(y) = e^{-q(y)} \left(\frac{q(y)}{a} + \frac{1}{2} - \frac{\Psi'(\theta_0)}{a} \right) \left[1 + \mathcal{O}\left(\frac{(\log Q)^2}{Q^2}\right) \right]. \quad (42)$$

The omitted R -sector is beyond all powers of $1/Q$; after substitution of the saddle scale it is of order $\exp\{-2t(y)\}$.

Proof. Write $t = e^q$. From (35),

$$xt = \frac{q}{a} + \frac{1}{2} - \frac{\Psi'(\theta)}{a} + \mathcal{O}(te^{-2t}).$$

Also,

$$K''(t) = \frac{D(q)}{t^2} + \mathcal{O}(e^{-2t}), \quad D(q) = \frac{q}{a} + \mathcal{O}(1).$$

Substituting these relations into the logarithm of the leading saddle formula in [theorem 6.3](#) gives

$$Y = \frac{q^2}{2a} - \left(\frac{1}{a} + \frac{1}{2} \right) q + \frac{1}{2} \log q + C(\theta) + \mathcal{O}(q^{-1}). \quad (43)$$

Now seek

$$q = Q + c + \frac{d \log Q + e}{Q} + \mathcal{O}((\log Q)^2/Q^2).$$

Substitution into (43) first gives $c = 1 + a/2$, then $d = -a/2$, and finally $e = c^2/2 - aC(\theta_0)$. Replacing the exact phase by θ_0 changes only the stated remainder because Ψ is smooth. Formula (42) follows from (7). $\square$

Remark 7.2 (A numerically stable parametric algorithm). For extreme arguments, direct inversion of F is ill-conditioned. A stable algorithm is:

1. solve the scalar equation $K'(t) = 1 - x$ or its Lambert form;
2. evaluate $Y(t)$ from the saddle expansion;
3. invert the monotone map $t \mapsto Y(t)$;
4. recover x from (35).

All products are evaluated at positive t , avoiding cancellation in $F(x)$ itself.

8 Sharp polynomial reproduction on dyadic grids

This section uses the same zero multiplicities in a different way. Define the Appell polynomials A_n by

$$\frac{e^{zt}}{M(t)} = \sum_{n=0}^{\infty} A_n(z) \frac{t^n}{n!}. \quad (44)$$

Equivalently,

$$A_n(z) = M(D)^{-1} z^n, \quad D = \frac{d}{dz}, \quad (45)$$

where the operator series terminates on every polynomial. Its coefficients are Bell polynomials in the negatives of the Bernoulli cumulants (19). For example,

$$A_0 = 1, \quad A_1 = z, \quad A_2 = z^2 - \frac{1}{9}, \quad A_3 = z^3 - \frac{z}{3}, \quad A_4 = z^4 - \frac{2}{3}z^2 + \frac{31}{675}.$$

Theorem 8.1 (Dyadic Appell reproduction). *Fix $r \geq 0$ and $h = 2^{-r}$. For every $0 \leq n \leq r$,*

$$\boxed{x^n = h \sum_{k \in \mathbb{Z}} A_n(kh) u(x - kh).} \quad (46)$$

Consequently every polynomial p of degree at most r satisfies

$$p(x) = h \sum_{k \in \mathbb{Z}} [M(D)^{-1} p](kh) u(x - kh). \quad (47)$$

For each fixed x , the sums are finite because u is compactly supported.

Proof. Consider the exponential generating function of the right-hand side:

$$G(t, x) = \frac{h}{M(t)} \sum_{k \in \mathbb{Z}} e^{tkh} u(x - kh).$$

Poisson summation applied to $y \mapsto e^{-ty} u(y)$ gives

$$G(t, x) = e^{tx} \left[1 + \sum_{m \neq 0} \frac{M(t + i\omega_m)}{M(t)} e^{i\omega_m x} \right], \quad \omega_m = \frac{2\pi m}{h}. \quad (48)$$

Now $\omega_m = 2\pi m 2^r$. By [theorem 3.1](#), $M(t + i\omega_m)$ has a zero at $t = 0$ of order

$$1 + \nu_2(m 2^r) = r + 1 + \nu_2(m) \geq r + 1.$$

Therefore the alias sum contributes nothing through degree t^r , and comparison of coefficients in (48) proves (46). Linearity gives (47). $\square$

The theorem specializes immediately to Legendre, Lagrange, Bernoulli, Bell, or any other polynomial family. The exact first failure beyond the reproduction threshold is the useful sharp endpoint of this presentation; it overlaps the repository's existing Appell, synthesis, and mesh-sharpness development and is not claimed here as a standalone novelty result.

For odd m , define

$$T_m = \prod_{\ell=1}^{\infty} \operatorname{sinc}\left(\frac{\pi m}{2^\ell}\right). \quad (49)$$

Theorem 8.2 (Exact first aliasing defect and sharpness). *With $h = 2^{-r}$,*

$$\begin{aligned} h \sum_{k \in \mathbb{Z}} A_{r+1}(kh) u(x - kh) &= x^{r+1} \\ &= -(r+1)! \sum_{\substack{m \in \mathbb{Z} \setminus \{0\} \\ m \text{ odd}}} \frac{T_m}{(i 2\pi 2^r m)^{r+1}} e^{2\pi i m 2^r x}. \end{aligned} \quad (50)$$

The right-hand side is not identically zero. Hence degree r is sharp: the shift-invariant space generated by $u(\cdot - kh)$ has Strang–Fix polynomial reproduction degree exactly r .

Moreover, as $r \rightarrow \infty$, the $m = \pm 1$ pair dominates the normalized defect uniformly; the remaining harmonics are $\mathcal{O}(3^{-(r+1)})$ relative to the first pair.

Proof. In (48), only those m for which the zero order is exactly $r+1$ can contribute to the coefficient of t^{r+1} ; these are the odd integers. Put $\omega_m = 2\pi 2^r m$. The first $r+1$ factors of $M(t + i\omega_m)$ vanish linearly. Their product has leading term

$$-\frac{t^{r+1}}{(i\omega_m)^{r+1}},$$

because the sum of the relevant integer phases is $m(2^{r+1} - 1)$, which is odd. The remaining factors equal T_m . Since $M(t) = 1 + \mathcal{O}(t^2)$, extracting the coefficient of t^{r+1} gives (50).

The $m = 1$ Fourier coefficient is nonzero because every factor in T_1 is positive, proving sharpness. Finally $|T_m| \leq 1$, and the absolute contribution of $|m| \geq 3$ contains the factor $|m|^{-(r+1)}$; comparison with $T_1 > 0$ proves first-harmonic dominance. $\square$

Corollary 8.3 (Defect for any degree- $r+1$ polynomial). *If $p(x) = \alpha x^{r+1} + \text{lower-degree terms}$, then the error in (47) is α times (50). In particular, for P_{r+1} , the multiplier is*

$$\alpha = 2^{-(r+1)} \binom{2r+2}{r+1}.$$

9 An integer-base family of atomic functions

The spectral mechanism is not peculiar to base 2. Let $b \geq 2$ be an integer and define

$$X_b = \sum_{n=1}^{\infty} \frac{b-1}{b^n} U_n, \quad \Phi_b(z) = \prod_{n=1}^{\infty} \operatorname{sinc}\left(\frac{(b-1)z}{b^n}\right). \quad (51)$$

The weights sum to 1, so $X_b \in [-1, 1]$. The same Fourier-decay argument gives an even C^∞ density u_b .

Let

$$\nu_b(m) = \max\{j \geq 0 : b^j \mid m\}, \quad c_b = \frac{\pi b}{b-1},$$

and let $s_b(m)$ be the sum of the base- b digits of m .

Theorem 9.1 (Base- b digital spectrum). *The following identities hold:*

$$\Phi_b(z) = \prod_{m \geq 1} \left(1 - \frac{z^2}{c_b^2 m^2}\right)^{1+\nu_b(m)}, \quad (52)$$

$$\mathcal{Z}_b(s) := \sum_{m \geq 1} \frac{1 + \nu_b(m)}{(c_b m)^s} = c_b^{-s} \frac{\zeta(s)}{1 - b^{-s}}, \quad (53)$$

$$N_b(c_b M) = \frac{bM - s_b(M)}{b-1}. \quad (54)$$

Thus the complete base- b digit sum is recoverable from the zero count:

$$s_b(M) = bM - (b-1)N_b(c_b M). \quad (55)$$

Proof. Euler's sine product and the substitution $m = b^{n-1}k$ give (52). The same divisor-indicator sum as before gives (53). Finally,

$$N_b(c_b M) = M + \sum_{j \geq 1} \left\lfloor \frac{M}{b^j} \right\rfloor = M + \frac{M - s_b(M)}{b-1},$$

which is (54). □

Corollary 9.2 (Base- b cumulants). *Odd cumulants vanish, and*

$$\kappa_{2r}^{(b)} = \frac{2^{2r-1} B_{2r}}{r} \frac{(b-1)^{2r}}{b^{2r}-1} = (-1)^{r+1} \frac{(2r)!}{r} \mathcal{Z}_b(2r). \quad (56)$$

In particular, all moments are rational for integer b .

For the N -factor truncation,

$$\mathcal{Z}_{b,N}(s) = c_b^{-s} \zeta(s) \frac{1 - b^{-Ns}}{1 - b^{-s}} = c_b^{-s} \zeta(s) [N]_{b^{-s}}, \quad (57)$$

so the q -integer phenomenon is universal across integer bases.

9.1 Base- b log-periodicity

Set $T = (b-1)t$, $a_b = \log b$, $q = \log T$, and $\theta = \{q/a_b\}$. The split-sum proof of [theorem 5.1](#) gives the following exact analogue.

Theorem 9.3 (Base- b transseries). *For $T \geq 1$,*

$$\begin{aligned} K_b(t) &:= \log \mathbb{E} e^{tX_b} \\ &= t - \frac{q^2}{2a_b} + \left(\frac{1}{2} - \frac{\log 2}{a_b} \right) q + \Psi_b(\theta) + R_b(T), \end{aligned} \quad (58)$$

where

$$\begin{aligned} \Psi_b(\theta) &= -\frac{b^\theta}{b-1} + \theta \log 2 + \frac{a_b}{2}(\theta^2 - \theta) \\ &\quad + \sum_{j \geq 0} \log(1 - e^{-2b^{j+\theta}}) + \sum_{k \geq 1} \log \left(\frac{\sinh(b^{\theta-k})}{b^{\theta-k}} \right), \end{aligned} \quad (59)$$

$$\begin{aligned} R_b(T) &= -\sum_{\ell \geq 0} \log(1 - e^{-2b^\ell T}) \\ &= \sum_{n \geq 1} \frac{b^{\nu_b(n)+1} - 1}{(b-1)n} e^{-2nT}. \end{aligned} \quad (60)$$

The meromorphic frequencies are $2\pi i k / \log b$.

Proof. Repeat the finite/tail split with $K_b(t) = \sum_{n \geq 1} f(T/b^n)$. The geometric linear sum contributes $t - b^\theta/(b-1)$; the logarithmic finite sum contributes

$$-\frac{q^2}{2a_b} + \left(\frac{1}{2} - \frac{\log 2}{a_b} \right) q + \theta \log 2 + \frac{a_b}{2}(\theta^2 - \theta).$$

The remaining reindexing is identical to the dyadic proof. Expanding the last product and grouping $n = mb^\ell$ gives (60). □

Remark 9.4. For noninteger $b > 1$, the random series and log-periodic asymptotics still make sense, but the exact digit-sum interpretation disappears. Integer radices are precisely where analytic scale invariance and arithmetic divisibility lock together.

10 Further rigorous consequences

10.1 Optimal sub-Gaussianity

The random-series model gives a useful global inequality.

Proposition 10.1 (Exact sub-Gaussian proxy). *For all real t ,*

$$M(t) \leq \exp(t^2/18). \quad (61)$$

The variance proxy $1/9$ is optimal. Consequently,

$$m_{2n} \leq \min\left(1, \frac{2^{n+1}n!}{9^n}\right). \quad (62)$$

Proof. The elementary inequality $\sinh y/y \leq e^{y^2/6}$ gives

$$M(t) \leq \exp\left(\frac{t^2}{6} \sum_{n \geq 1} 4^{-n}\right) = \exp(t^2/18).$$

No smaller Gaussian proxy is possible, because $K''(0) = \text{Var } X = 1/9$. Chernoff's argument gives $\Pr\{|X| \geq s\} \leq 2 \exp(-9s^2/2)$. Integration of this tail bound,

$$\mathbb{E}|X|^{2n} = 2n \int_0^\infty s^{2n-1} \Pr\{|X| \geq s\} ds,$$

gives $2^{n+1}n!/9^n$; compact support gives the competing bound 1. $\square$

10.2 Log-concavity

Each finite density u_N is a convolution of interval indicators and is therefore log-concave. Passage to the locally uniform limit gives:

Proposition 10.2. *The Rvachev density u , and more generally every u_b , is log-concave on its support.*

Strict log-concavity is more delicate because every finite stage has spline facets. It is isolated below as a conjecture.

10.3 A distributional smoothing/deconvolution calculus

From (7),

$$u = u_N * [2^N u(2^N \cdot)]. \quad (63)$$

For a test function φ , Taylor expansion against the scaled tail gives the distributional asymptotic

$$u \sim \sum_{j \geq 0} \frac{m_{2j}}{(2j)! 2^{2Nj}} D^{2j} u_N. \quad (64)$$

Conversely,

$$\widehat{u}_N(z) = \frac{\Phi(z)}{\Phi(z/2^N)}$$

gives

$$u_N \sim \sum_{j \geq 0} \frac{A_{2j}(0)}{(2j)! 2^{2Nj}} D^{2j} u, \quad (65)$$

again distributionally, with the reciprocal-moment coefficients generated by $1/M$. These expansions provide systematic high-order corrections to finite Thue–Morse splines without recomputing the 2^N -term formula (6).

11 Conjectures

The conjectures are ordered from concrete to structural.

Conjecture 11.1 (One-wave topology of the periodic fluctuation). *The dyadic function Ψ has exactly one maximum and one minimum on $\mathbb{R}/\mathbb{Z}$, both nondegenerate. More generally, Ψ_b has this property for every integer $b \geq 2$.*

The exponential gap between c_1 and the higher coefficients makes this highly plausible and numerically testable. A proof should follow from an explicit domination inequality for $\sum_{|k| \geq 2} k c_k e^{2\pi i k \theta}$ by the first harmonic.

Conjecture 11.2 (Strict log-concavity). *For every real $b > 1$, the density u_b is strictly log-concave on $(-1, 1)$:*

$$(\log u_b)''(x) < 0, \quad -1 < x < 1.$$

The non-strict statement is proved above. Strictness would sharpen concentration bounds and imply uniqueness properties for tilted saddle measures.

Conjecture 11.3 (Complete endpoint transseries). *After the exact decomposition of K_b , the saddle expansion for $u_b(1 - x)$ is Gevrey-1 in $1/\log t$ and Borel summable along the positive saddle. Its only beyond-all-orders exponential scales are those generated by*

$$e^{-2n(b-1)t}, \quad n \geq 1,$$

with coefficients determined by the base- b valuation sequence in (60); there are no hidden exponential actions.

Conjecture 11.4 (Sharp Legendre logarithmic scale). *For the nonzero even Legendre coefficients,*

$$\log |a_{2n}| = -\frac{(\log n)^2}{2 \log 2} + \mathcal{O}(\log n),$$

and, after subtraction of an explicit linear-log term, the upper envelope has a 1-periodic modulation in $\log_2 n$ governed by the same complex dimensions χ_k .

The caveat “upper envelope” is essential because the zero divisor of Φ can create deep arithmetic dips.

Conjecture 11.5 (Asymptotic exactness of the first alias wave). *For standard Sobolev and uniform norms, the optimal degree- $r + 1$ polynomial quasi-interpolation error on the grid $2^{-r}\mathbb{Z}$, after its natural normalization, is asymptotic to the $m = \pm 1$ wave in (50); no alternative local coefficient functional can cancel that wave without sacrificing stability.*

The explicit defect proves first-wave dominance for the canonical Appell scheme. The conjecture asserts its minimax optimality across stable local schemes.

12 Promising research directions

12.1 Rigorous complex-dimension calculus

The spectral zeta (11) is simple enough to support a complete “explicit formula” theory parallel to that of self-similar strings. Natural next objects include:

$$\sum_m (1 + \nu_2(m)) g(2\pi m/T), \quad \log \det_\zeta(\lambda^2 + \rho_m^2),$$

and smoothed zero-counting errors. Their residues should give a unified family of periodic functions, with exact transfer formulas between test functions.

12.2 A fully rigorous endpoint local limit theorem

Theorem 6.3 can be strengthened into a theorem uniform in complex sectors by treating the exponentially tilted summands as a triangular array. The useful target is an explicit remainder with computable constants:

$$\left| \frac{F(x)\sqrt{2\pi K''(t)}}{e^{K(t)-t(1-x)}} - \sum_{j=0}^{m-1} C_j(t) \right| \leq C_m(\log t)^{-m}.$$

This would turn the asymptotic formulas into certified inverse-Fabius algorithms.

12.3 Interval-certified inverse evaluation

The phase-locked Lambert equation is well suited to interval Newton methods. A rigorous implementation would:

1. bracket t with W_{-1} ;
2. bound Ψ , Ψ' , and Ψ'' by truncating (32) with an analytic tail estimate;
3. bound R by its positive series (30);
4. propagate the enclosure through the saddle expansion.

This is likely far more efficient than direct evaluation of a flat C^∞ function near zero.

12.4 Legendre and other orthogonal bases

The Fourier–Bessel bridge (25) converts the problem of sharp Legendre coefficients into a turning-point integral involving a lacunary canonical product. A combined Debye–Airy analysis of j_n and the digital envelope of Φ may prove [conjecture 11.4](#). Analogous transforms exist for Gegenbauer, Chebyshev, and Hermite systems.

12.5 q -binomial refinement of finite cascades

Equation (21) identifies the first q -object: $[N]_{2^{-s}}$. Higher exterior or subset statistics of the finite divisor should produce Gaussian binomial coefficients $\begin{bmatrix} N \\ k \end{bmatrix}_{2^{-s}}$. A concrete program is to define the k -fold distinct-zero spectral sums

$$E_{k,N}(s) = \sum_{1 \leq n_1 < \dots < n_k \leq N} 2^{-s(n_1 + \dots + n_k)} = 2^{-sk(k+1)/2} \begin{bmatrix} N \\ k \end{bmatrix}_{2^{-s}},$$

then identify the corresponding cumulant or spline operators. This would connect the repository's q -binomial formulas to the spectral divisor rather than only to coefficient algebra.

12.6 Digit masks beyond uniform boxes

Replace sinc by the Fourier transform of a compactly supported mask and form

$$\prod_{n \geq 1} P(z/b^n).$$

When the zeros of P fall in finitely many rational arithmetic progressions, the resulting spectral zeta should be a finite linear combination of terms

$$L(s, \chi) \frac{Q(b^{-s})}{1 - b^{-s}},$$

with Dirichlet L -functions and rational Q . This offers a route from Thue–Morse to Rudin–Shapiro and more general automatic sequences.

12.7 Formal verification

The repository already contains formal work on zero multiplicity, analytic zero order, weighted spectral zeta identities, periodic corrections, Lambert-phase algebra, and polynomial-filter sharpness. For this report, the useful next task is an exact crosswalk for:

- (i) the divisor regrouping in [theorem 3.1](#);
- (ii) the identity $N(2\pi M) = 2M - s_2(M)$;
- (iii) the finite spectral q -integer (21);
- (iv) the Bell-polynomial cumulant formula;
- (v) coefficient extraction in the Appell reproduction theorem.

Relevant declarations may cover ingredients or nearby formulations, but this manuscript does not establish exact statement identity. The Mellin-contour and saddle arguments in particular must be treated as paper proofs unless a named declaration with matching assumptions and conclusion is recorded.

13 Conclusions

The organizing viewpoint of this manuscript is that the dyadic sinc product carries an arithmetic spectral divisor:

$$\text{mult}_{2\pi m} \Phi = 1 + \nu_2(m).$$

From this one line follow:

- the spectral zeta $\zeta(s)/((2\pi)^s(1 - 2^{-s}))$;
- the exact digit-sum count $2M - s_2(M)$ and spectral Thue–Morse parity;
- Bernoulli cumulants as reciprocal zero power sums;
- complex dimensions $2\pi i k / \log 2$;
- an exact periodic-plus-exponential transseries for $\log M(t)$;
- a phase-corrected Lambert- W_{-1} endpoint saddle;
- a refined inverse-Fabius asymptotic;
- and a sharp dyadic hierarchy of polynomial reproduction and aliasing.

The integer-base extension shows that this is not an accidental collection of dyadic tricks. It is a general mechanism in which scale invariance, valuation arithmetic, Mellin poles, and flat compactly supported densities are different manifestations of the same structure. This conclusion is a synthesis of manuscript arguments with substantial current-corpus overlap, not a claim of priority or of complete Lean formalization.

A A compact derivation of the Mellin kernel

For completeness, for $0 < c < 1$,

$$\log(1 + x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi}{s \sin \pi s} x^s ds.$$

For $0 < x < 1$, closing to the right and accounting for orientation gives

$$-\sum_{n \geq 1} \operatorname{Res}_{s=n} \frac{\pi x^s}{s \sin \pi s} = -\sum_{n \geq 1} \frac{(-1)^n}{n} x^n = \log(1+x).$$

Analytic continuation gives the identity for all positive x . Substitution $x = t^2/(2\pi m)^2$, followed by the absolutely convergent sum over the zero divisor, yields (5.1).

B Coefficient generator for arbitrary saddle order

Let

$$\Lambda_j = \frac{K^{(j)}(t)}{j! K''(t)^{j/2}}, \quad j \geq 3.$$

Introduce a bookkeeping parameter ε by assigning weight $j - 2$ to Λ_j :

$$\mathfrak{S}(\varepsilon) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-z^2/2} \exp\left(\sum_{j \geq 3} \varepsilon^{j-2} \Lambda_j(\mathrm{i}z)^j\right) dz.$$

Then the coefficient of ε^{2m} is the m -th relative saddle correction. Expanding the exponential gives the explicit partition formula

$$C_m(t) = \sum_{\substack{k_3, k_4, \dots \geq 0 \\ \sum_{j \geq 3} (j-2)k_j = 2m \\ \sum jk_j \text{ even}}} \frac{(-1)^{\frac{1}{2} \sum jk_j} (\sum jk_j - 1)!!}{\prod_{j \geq 3} k_j!} \prod_{j \geq 3} \Lambda_j^{k_j}.$$

Only finitely many indices occur at each m . This is the Bell-polynomial form of the arbitrary-order expansion requested in symbolic applications.

C Repository-audit reproducibility

The archive contains:

- `repository_audit.md`, with the dated corpus scope, deterministic aggregate inventory, audit method, and targeted overlap register;
- `numerical_experiments.py`, with all high-precision checks and figure generation;
- `numerical_results.tex`, generated by that program;
- `numerical_run.txt`, with the successful command, environment, and generated-output hashes; and
- the three figures used in this report.

The audit is intentionally bounded: deterministic inventory and targeted source/declaration searches establish overlap, but automated text search does not decide mathematical equivalence. The immutable arrival ledger records the received bundle; repaired live-file hashes belong to the repository's separate live ledger.

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